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Fuzzy Logic Combined Logistic Regression Methodology for Gas Turbine First-Stage Nozzle Life Prediction

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Abstract. Significant aspects of intelligent maintenance include the abilities to diagnose impending failures, prognose the remaining useful lifetime of the process and schedule maintenance operations so that uptime is maximized. Prognosis is probably the most difficult of the three issues leading to total intelligent maintenance. This paper describes a fuzzy logic combined logistic regression method of fatigue severity assessment and remaining useful life prediction of gas turbine hot components. Logistic regression method is proposed to derive fuzzy logic rule base using historical maintenance running records and engineers' experience. Implementation of the prognostic methodology presents a great opportunity to significantly enhance current engine health monitoring capabilities and risk management practices.

Introduction

Maintenance of hot components such as combustors, nozzles is a major concern in gas turbine operation. Hot components are generally made of expensive superalloy in order to withstand severe operational environment, however, the lives of hot components are relatively shorter than other ones, such as rotor and casing. Overhauls of critical parts are conducted with certain intervals to inspect degradation severity, furthermore repairs and replacements might take place according to the degree of the degradation. Scheduled downtime for hot component inspections takes a large percentage of total downtime [1]. Life management of hot components is important for saving the operation and maintenance cost and improving the availability of gas turbine.

First-stage nozzles in turbine section are located right next to combustors and exposed to hottest gases in turbine components. Cracking is one of the most frequent failure modes for nozzle [2], which might even cause fatal failure of gas turbines. Several approaches such as fracture mechanics method [3, 4], trend analysis [2], etc., have been studied for predicting the crack growth behavior in hot components. Fracture mechanics method is a fundamental approach to accomplish a reasonable prediction [3, 4] by considering multi-inputs such as temperature, stress and structural analysis using finite element analysis.

So far investigations on life prediction using routine inspection data are not sufficient. In order to conduct life prediction dynamically to assist traditional mechanics life analysis, inspection data is becoming more important. Usually, there are bunch of inputs representing component degradation severity, for example, crack length, number of cracks, eroded area, which should be considered for an overall damage assessment of components. In this paper, a fuzzy logic and based method is proposed to evaluate and predict remaining life by fuse multiple inputs, where logistic regression is employed to determine parameters in fuzzy model. The proposed methodology has been applied to the inspection data of first stage nozzles collected from a Hitachi gas turbine for crack growth analysis, the life prediction results demonstrate the effectiveness of the fuzzy logic based prognostic methodology.

Fuzzy Logic Combined Logistic Regression Based Prognostic Methodology

In this context, Takagi-Sugeno model is employed, which is a nonlinear system model represented by fuzzy rules of the type

$$R^i : \text{If } x_1 \text{ is } A_1^i \text{ and } \dots \text{ and } x_m \text{ is } A_m^i, \text{ then } y^i = g(x_1, \dots, x_m) \quad (1)$$

where R^i ($i=1,2,\dots,n$) denotes the i -th fuzzy rule, x_j ($j=1,2,\dots,m$) are input variables and y^i is an output. Furthermore, and $A_1^i, A_2^i, \dots, A_m^i$ are the linguistic values taken by the input variables in the antecedent (if-part) of the i -th fuzzy rule. The meaning of these linguistic values is defined by corresponding membership functions. $g(\bullet)$ is the function that implies the value of y when $x_1, \dots, x_m$ satisfies the premise. Where y is an index to represent probability of failure in terms of a vector of inputs $x_1, \dots, x_m$. In this paper, we proposed to put $g(\bullet)$ into logistic regression format to meet the probability of failure requirement.

$$g(x_1, \dots, x_m) = \frac{1}{1 + \exp(-(a_0 + a_1 x_1 + \dots + a_m x_m))} \quad (2)$$

As shown in Eq.s (1) and (2), this fuzzy model describes a nonlinear input-output relation. The overall output of a T-S fuzzy model is given as

$$\hat{y} = \frac{\sum_{i=1}^n \beta^i y^i}{\sum_{i=1}^n \beta^i} \quad (3)$$

$$\beta^i = \prod_{j=1}^m \mu_{A_j^i}(x_j) \quad (4)$$

Where $\mu_{A_j^i}$, is the membership function defining the meaning of the linguistic value A_j^i . a_j^i are the parameters contained in the consequent (then-part) of the i -th fuzzy rule.

The identification of a T-S fuzzy model, using input-output data, is divided into two steps: structure identification and parameter estimation. The former consists of antecedent structure identification and consequent structure identification. The latter consists of antecedent and consequent parameter estimation where the consequent parameters are the coefficients of the linear expressions in the consequent of a fuzzy rule. Generally, the number of inputs is the number of features extracted from sensor readings / inspection records, the parameters could be calculated using maximum likelihood method [5].

Application

In this paper, the inspection data of a 25 MW Hitachi gas turbine (H25), acquired over a period of four years (1990-1993) was collected to validate the method proposed. Specifically, data of 57 nozzles used in a gas turbine was used for degradation assessment and life prediction. Based on the maintenance policy to the gas turbine we referred to, the inspection interval was one year, approximately 7,000 operation hours. The H25 gas turbine is with 42 first-stage nozzles. During the four years operation, 57 nozzles' data were obtained since the damage rate of each nozzle is diverse, therefore some of the nozzles (for example, 15 in this case) were replaced during yearly periodical inspections.

Fig.1 shows the fuzzification phase of three inputs as maximum crack length, total crack length and operating hours used in this research. Maximum crack length represents the length of the longest crack in a nozzle. Total crack length refers to the sum of the length of all cracks observed on a nozzle. Each fuzzy variable is divided into five fuzzy subsets representing fatigue severity as good, slight, moderate, heavy and unacceptable correspondingly according to engineers' experience. Classifications of these subsets were determined based on inspection data.

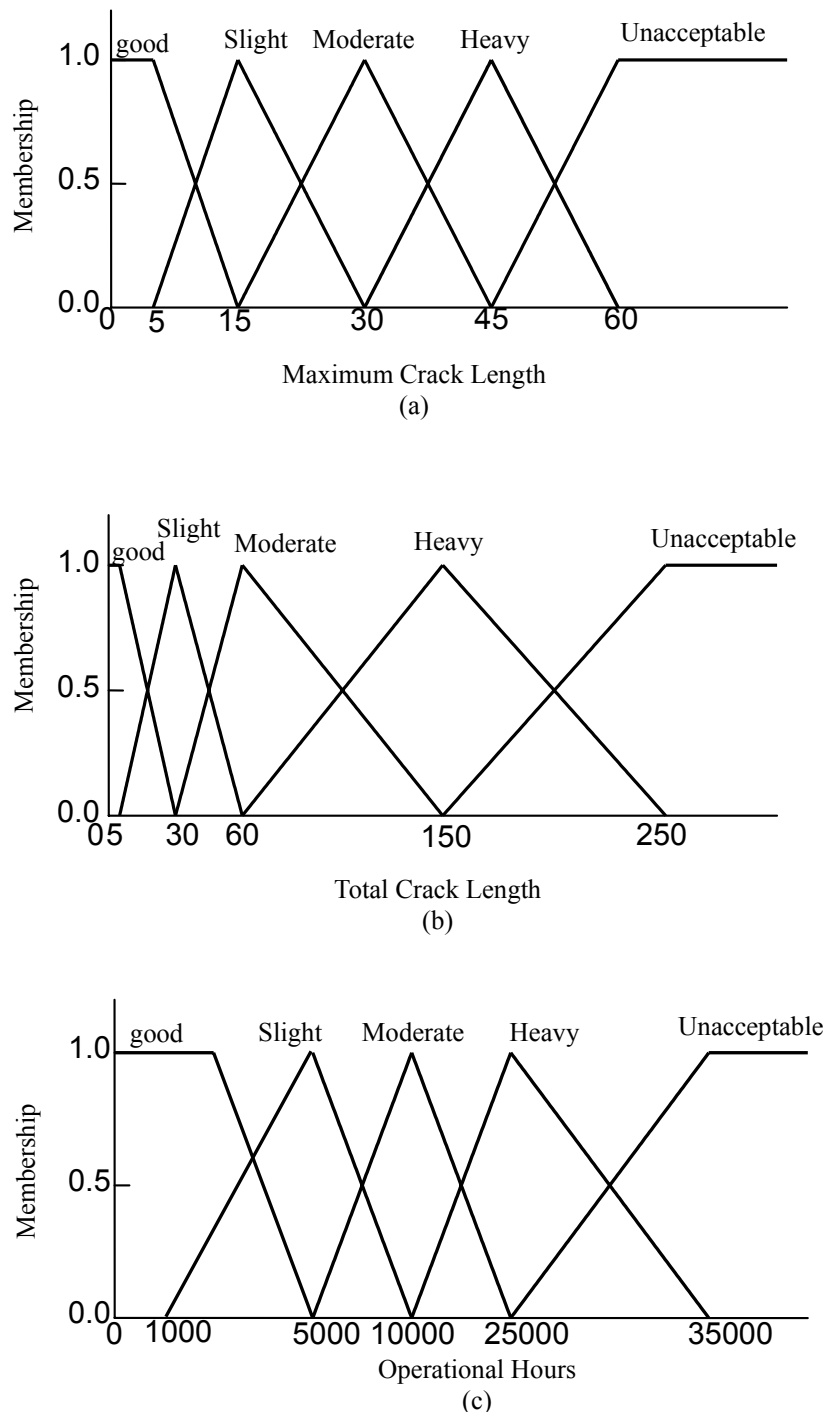


Fig.1 Fuzzy subsets of inputs

In this case, inspection data of 21 among the total 57 nozzles was used for training, to determine the coefficients in equation 2. The rest inspection data set of 36 nozzles was used for verification. The prediction results of one of the nozzles (nozzle #7) are shown in figure 2. Unacceptable line is defined according to technicians' experience. The solid line represents actual running condition, this nozzle was replaced at the end of the 4th year. At the end of first year, based on one year records,

the prediction trend is shown in dash line 'Prediction 1', which evaluates the degradation severity of the following years, shows that the nozzle could be used safely in the second and third year, but the fourth year's performance is not acceptable. The prediction conducted at the end of second year, see the dash line marked as 'Prediction 2', shows the nozzle can work safely until the end of the 4th year, but can not be used for the fifth year, which matches the nozzle's actual life. Similarly, 'Prediction 3' which was derived using first three years' inspection data shows that the nozzle can be used for next year, but is not applicable for the 5th year. At the end of fourth year, the next one step prediction crosses the unacceptable line, which means the nozzle cannot be used for another inspection round. The prediction results of all 36 vanes match the actual situations well. Fuzzy logic combined logistic regression prognostic method provides technicians with the information that most probably when the nozzle cannot fulfill its mission, this is very important for near-zero downtime running, spare parts ordering and optimal maintenance scheduling.

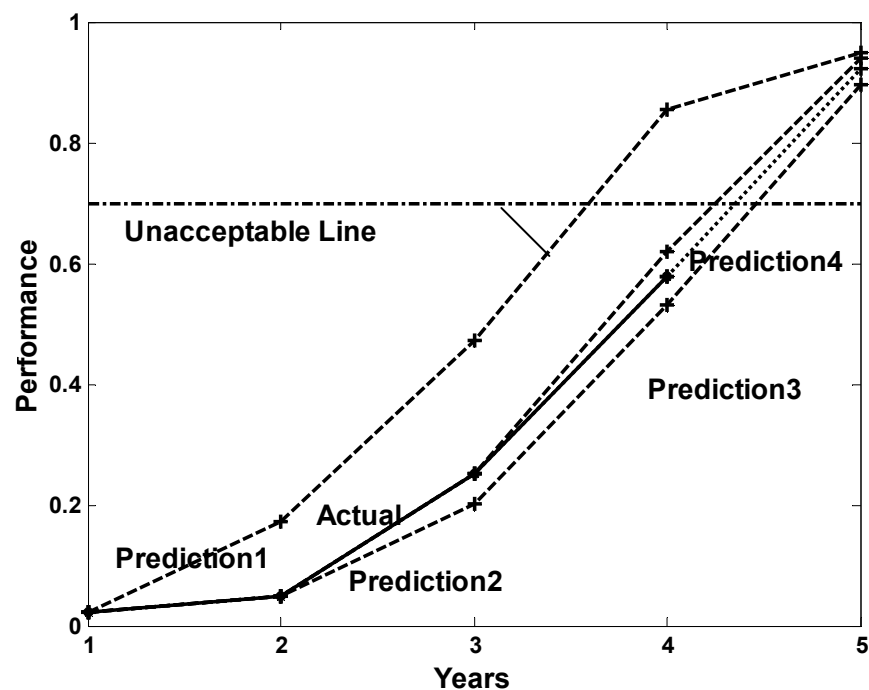


Fig.2 Life prediction result of nozzle #7

Obviously, the prediction results illustrate that there were some deviations in early stage of life prediction, the accuracy was gradually improved in the latter stage of life prediction. Therefore, the effectiveness of fuzzy logic based prognostic method was validated.

Conclusions

It is demonstrated that the fuzzy logic combined logistic regression approach presents a reasonable prediction for damage assessment of gas turbine nozzle. We focused on the damage caused by cracking in this study, some other damage modes such as erosion and creep deformation are critical as well and being taken into consideration by our team. Note that it is not easy to define the damage rate from history data via empirical judgement, since the relationship between features of damage such as cracking, erosion, and corrosion of components are interactional according to operating conditions, usage and other factors. Therefore, intelligent methodologies are feasible and promising to compensate traditional mechanics analysis of hot components life prediction.

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